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Successful Execution of World's First ZLD Pilot with Extreme Salinity Unconventional Frac Flowback Produced Water

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Abstract

A significant portion of the water used for the hydraulic fracturing of unconventional hydrocarbon resources is later recovered when subject wells are flowed back post fracturing. Treating fracturing flow-back produced water from unconventional resources unlocks its full value and provide alternative supplies of low salinity water to seawater desalination and ground water consumption, while promoting a circular water economy approach through 4Rs (Reduce, Reuse, Recycle, and Recovery).

The main objective of this work is to highlight the design and deployment of a zero-liquid discharge (ZLD) flowback produced water management solution to treat quite challenging extreme salinity unconventional fracturing flow back flowback produced water (up to 250,000 ppm Total Dissolved Solids, TDS), and generate the desired high-quality low-salinity water suited for fracking applications. The designed ZLD unit comprised of a unique combination of a pretreatment system to remove residual oil/ H₂S and dynamic vapor compression technology for salt removal from hypersaline flowback produced water. The pretreatment system relied on chemical scavengers to lower the dissolved H₂S from flowback produced water, while induced gas floatation with nitrogen gas was used to remove dispersed hydrocarbons. The pretreated flowback produced water is then processed in a dynamic vapor compression unit to effectively remove salts and generate low salinity "fresh water".

The ZLD pilot was successfully executed at one of the unconventional well pad site locations, wherein a primary separator separated the three flowing phases (water, oil, and gas) from producing wells. The separated flowback produced water contained more than 500 ppm H₂S, about 1,000 ppm of residual hydrocarbons, and an extremely high salinity (up to 250,000 ppm TDS). The pilot unit was sized to process about 220 barrels per day (BPD) of flowback produced water obtained from the separator to generate distillate (fresh water) and remove salts as concentrated brine. The results obtained from ZLD pilot were very promising. The system was able to remove residual hydrocarbons and dissolved H₂S to less than 1 ppm, and most importantly, lower the salinity of flowback produced water to less than 200 ppm TDS, generating fresh water at an average 70-80% recovery efficiency. The pilot was operated at a range of salinities from 50,000 ppm to 250,000 ppm TDS (with different blending ratios of flowback produced water and freshwater) to develop the ZLD operating envelope for a spectrum of flowback produced water salinities.

In summary, ZLD is a breakthrough industry-first pilot in the world that can promote flowback produced water recycle/reuse and circular water economy in unconventional resources operations. The impressive performance and engineering excellence achieved during the execution of this pilot presents a step-change in flowback produced water desalination and reuse, which has huge potential to revolutionize water treatment and desalination practices followed in the oil and gas industry.

Introduction

Considerable volumes of water are produced today after fracking operations from unconventional oil and gas resource globally. The flow back flowback produced water volumes from unconventional oil and gas wells are estimated to be in the range of 0.5 to 3.8 million gallons per well during the production from first 5-10 years and 20-50% of these volumes are generated from the first 6-months of production (Kondash et al. 2017). The handling and disposal of these large quantities of flowback produced water pose environmental concerns, that necessitate complex and resource-intensive operations to mitigate. In some cases, the disposal of flowback produced water volumes also requires costly disposal facilities for deep-well injection.

Paradoxically, hydraulic fracturing in unconventional shales also consumes significant volumes of fresh water. For example, fracturing a single well needs 7,000 to 21,000 m³ of fresh water on average as part of unconventional oil and gas development (O'Hern et al. 2021). Sourcing of water for fracking has proven to be a major challenge for oil & gas companies (Whitfield 2014). A significant portion of these volumes are recovered post-fracking and during flowback (Freyman, 2014).

Water is essential for socioeconomic development and is a contributing factor in nearly every United Nation's (UN) Sustainable Development Goal (SDG). Treating and recycling flowback produced water can unlock that value and provide alternative supplies to currently used non-renewable fresh water sources during fracturing operations in unconventional resources development. The adoption and wider practice of such sustainable option would also preserve the water supply for human/environmental needs besides promoting circular water economy through 4Rs (Reduce, Reuse, Recycle, and Recover).

Shafer et al. (2013) reviewed a number of flowback produced water-based desalination technologies for reuse of high-salinity shale gas flowback produced water and identified that mechanical vapor compression (MVC), membrane distillation (MD), and forward osmosis (FO) are the best suited technologies for desalination of high-salinity flowback produced water. Minier-Matar et al. (2014) evaluated membrane distillation process at the bench scale on flowback produced water samples obtained from an unconventional shale play (with a salinity of 47,761 ppm and 49 ppm total organic carbon/TOC) and demonstrated the salt rejection efficiency of 99.7% in the distillate (less than 145 ppm TDS). Pawar et al. (2022) conducted laboratory and pilot scale studies on membrane distillation for the desalination of flowback produced water from an unconventional reservoir. The results showed 50% water recovery efficiency with 99.7% salt ion rejection on 127,000 ppm Total Dissolved Solids (TDS) flowback produced water. Hsieh et al. (2023) demonstrated the performance of vacuum membrane-distillation (VMD) for desalinating high-salinity flowback produced water (104,000 ppm TDS) obtained from unconventional wells to generate fresh water (less than 150 ppm TDS). The vapor compression has been integrated with membrane distillation (MD) in this study to improve the energy efficiency of desalination process.

Flowback produced water often comes with contaminants such as dissolved H₂S, TSS, and dispersed oil. Fit-for-purpose pretreatment technologies (Ahmadun et al., 2009; Amakiri et al., 2022; Abdelhamid et al., 2025) should be used based on the characteristics of flowback produced water to effectively remove these contaminants to acceptable levels prior to the desalination of flowback produced water. Eyitayo et al. (2023) reviewed purpose-specific water treatment units used in various activities within the oil and gas industry and presented historical development of current water treatment practices including some technological advancements. The low salinity water obtained from flowback produced water desalination will be useful not only for fracking, but also for other applications such as crude oil desalting and reservoir re-injection

for improved/enhanced oil recovery (IOR/EOR). [Ahmed et al. \(2022\)](#) evaluated the potential of flowback produced water desalination to conserve the currently used ground water in process/utility water applications including the crude oil desalting. Some recent experimental studies also demonstrated the beneficial effects of low salinity water obtained from flowback produced water desalination in SmartWater flooding and different mobility control processes used for enhanced oil recovery ([Hoecherl et al. 2024](#); [Ayirala et al. 2024](#)).

Based on the review provided above, it becomes clear that there are not that many desalination pilot studies reported to date using unconventional frac flowback produced water. Furthermore, most of the reported studies in the existing knowledge are limited to flowback produced water salinities up to 120,000 ppm total dissolved solids (TDS) with moderate water recovery efficiencies in the range of 50-60%. This underscores the significance of developing and testing sustainable solutions to generate fresh water from extreme salinity and quite challenging unconventional frac flowback produced water streams (up to 250,000 ppm TDS), targeting water recovery efficiencies as high as 70%. The development of such a solution will extend the desalination envelope to cover the entire spectrum of flowback produced water salinities generated from unconventional oil and gas fields globally, in addition to significantly reducing the waste discharge volumes towards achieving the near-Zero Liquid Discharge (ZLD).

The objectives of this study are: (1) Design an integrated ZLD solution comprising of a customized pretreatment system and a desalination unit, (2) Test the integrated unit at an unconventional well pad site to desalinate the extreme salinity (up to 250,000 ppm TDS) unconventional frac flowback produced water and generate fresh water at 65-70% recovery efficiency, (3) Extend the applicability of the designed unit to process a range of brine salinities expected from unconventional oil and gas wells, and (4) Demonstrate the suitability of fresh water obtained from ZLD pilot unit in stimulation operations in unconventional wells.

ZLD Pilot Design Basis

The scope of this project involves conducting the field trial of ZLD water management solution comprising of dynamic vapor recovery/compression (DyVaR) technology for the treatment of frac flowback produced water. Successful field trial is necessary to demonstrate the capabilities of the ZLD solution to treat the hypersaline frac flowback produced water streams (up to 250,000 ppm TDS), generate the desired quality water needed for fracking operations, and subsequently qualify ZLD technology for upscaling evaluations.

For the treatment of water to remove sour gas (H_2S) and suspended hydrocarbons, a pretreatment setup has been custom designed with proven capabilities to achieve the required criteria for water processing through DyVaR.

The following data and key performance metrics were used as design basis for the ZLD skid:

- Product salinity < 500 ppm
- System recovery 65-70%
- Inlet flow rates up to 150 barrels per day (BPD)
- Feed salinity with TDS up to 250,000 ppm
- Feed temperature 95-140oF

ZLD Pilot System Description

The ZLD pilot unit is designed to treat flowback produced water with a pretreatment system and a DyVaR unit to process up to about 150 BPD, and generate a product water salinity of <500 ppm at 65-70% system recovery. The feed to the pretreatment system is the wastewater that has already completed a primary oil-

water separation via a three-phase separator at an unconventional well pad site, with a salinity up to 250,000 ppm TDS, OiW (oil in water) < 1,000 ppm and more than 500 ppm H₂S.

Pretreatment System

Pretreatment unit has been designed to remove H₂S and dispersed hydrocarbons, from process feed water, before sending the feed water into the DyVaR. The main goals of the pretreatment unit are: (1) Scavenge H₂S in flowback produced water to less than 10 ppm, (2) Remove suspended hydrocarbons to the DyVaR pilot unit's acceptable operating limit, and (3) Safely vent H₂S free effluent away from the operating area. The process flow schematic of the pretreatment unit is shown in Figure 1.

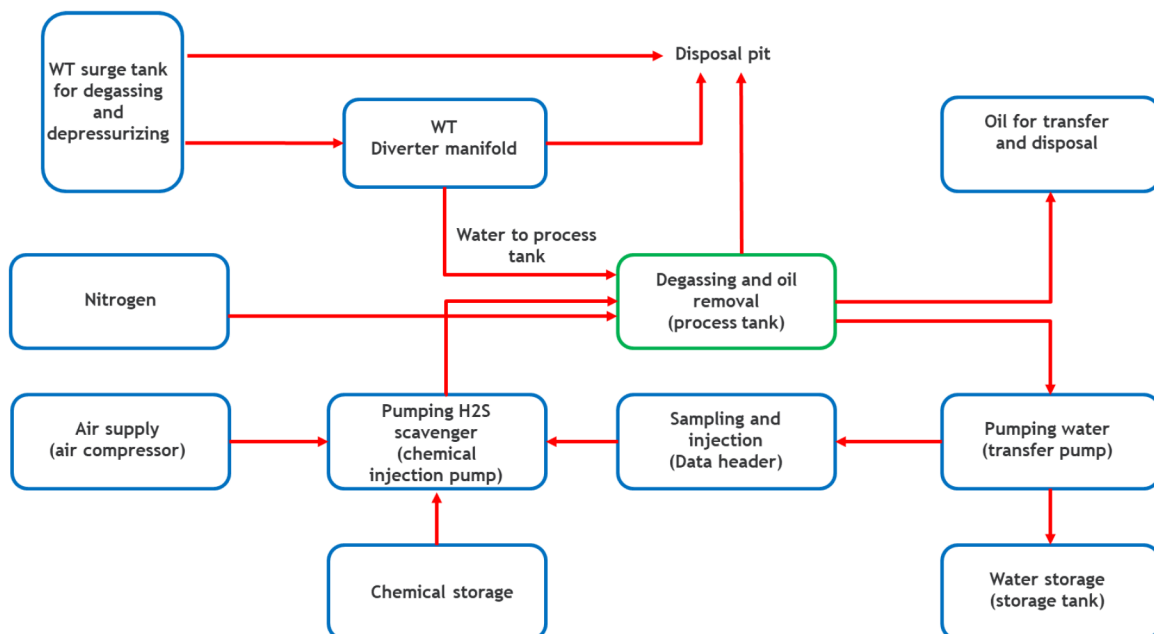


Figure 1—Process flow schematic of the pretreatment system

As seen in Figure 1, the pretreatment system mainly relies on the following steps to effectively scavenge H₂S and remove dispersed oil from the frack flowback flowback produced water.

1. Flowback produced water from well testing (WT) Separator was first collected in WT surge tank and vented.
2. A 250 bbl, 85 psig process tank was used to receive the flowback produced water from the WT surge tank (62.5bbls × 2 batches = 125 bbls for each pre-treatment batch) at transfer pressures less than 50psig.
3. H₂S gas released from the water exit the process tank to the flare pit/gas scrubber.
4. When the process tank is filled to 50%, the flowback produced water for WT process is diverted to the flare pit and the process tank feed valve is closed.
5. N₂ is pumped at 365 SCF from an air compressor through a 1" SS perforated pipe grid at the bottom of the process tank.
6. N₂ bubbles rise through the water in the tank with the intent of pushing out dissolved gases and oil to the top of the liquid layer.
7. The oil layer (if any) is detected by the interface level sensor.
8. Using the interface level sensor, the oil and gas free water is displaced from the process tank. The oil layer is removed from the process tank once the layer thickness is 0.25" via 2" hose to the oil disposal tank.

9. The water starts circulating through the process tank using a circulation pump connecting the outlet.
10. To reach the desired H_2S level, the scavenger injection pump is connected to the recirculation line where it can be pumped during the circulation of the water.
11. Once the water H_2S level reaches the desired level (below 10 ppm) which can be monitored through the data header installed on the re-circulation line of the process tank, the water is transferred to the storage tank (500 bbls) for feeding the pretreated water to the DyVaR desalination unit.

Dynamic Vapor Recovery/Compression Desalination Unit:

Dynamic Vapor Recovery (DyVaR) has been selected as a potential technology for evaluation in this study to desalinate the pretreated frac flowback produced water. The fluid entering the DyVaR system is heated to a temperature of approximately 80-90°C to achieve evaporation in cyclones and the evaporated water vapor is condensed to generate low salinity water. The detailed working principle of DyVaR unit is shown in Figure 2.

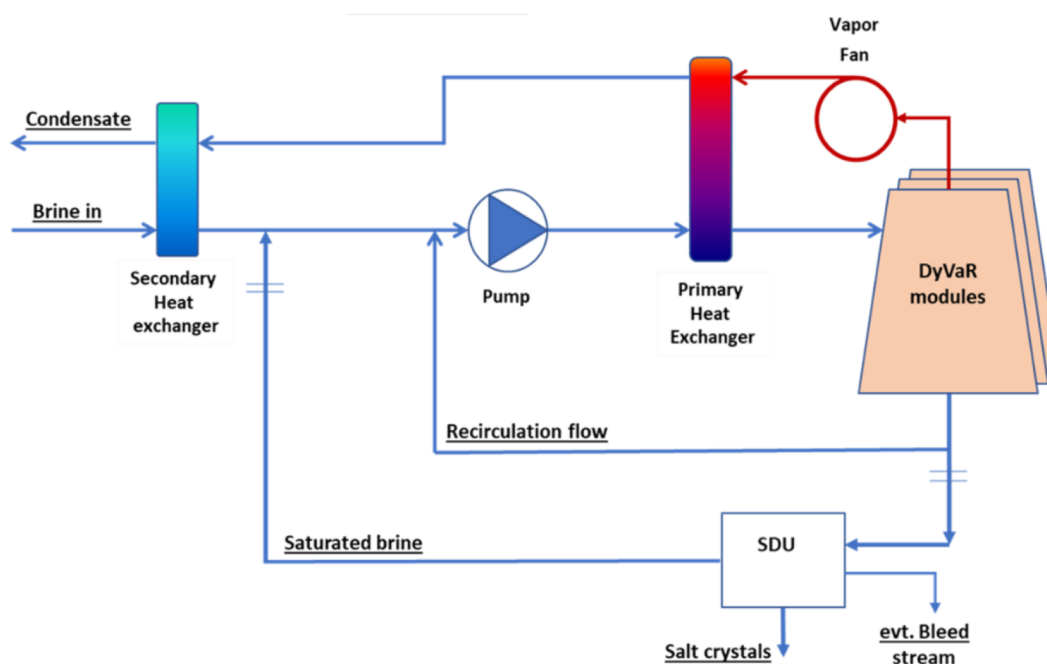


Figure 2—Detailed Principle of DyVaR Technology (Salttech, 2014)

DyVaR process is a low-heat cyclonic evaporation desalination process that is designed with heat recapture for optimal energy efficiency over a wider range of input water salinities. Influent water is heated partially through heat exchange with the distillate stream. Compressors apply a modest vacuum over a series of ceramic "cyclones", which facilitate the evaporation and distillation. The mother liquor is cycled back through the process via a recirculation pump, from which the brine discharge is purged and undistilled water is pumped back into the multi-cyclone evaporation unit. The result is typically 80+% water recovery (by volume) with ZLD reject slurry that can be further processed for mineral recovery. The flow schematic representation of the process is shown in Figure 3.

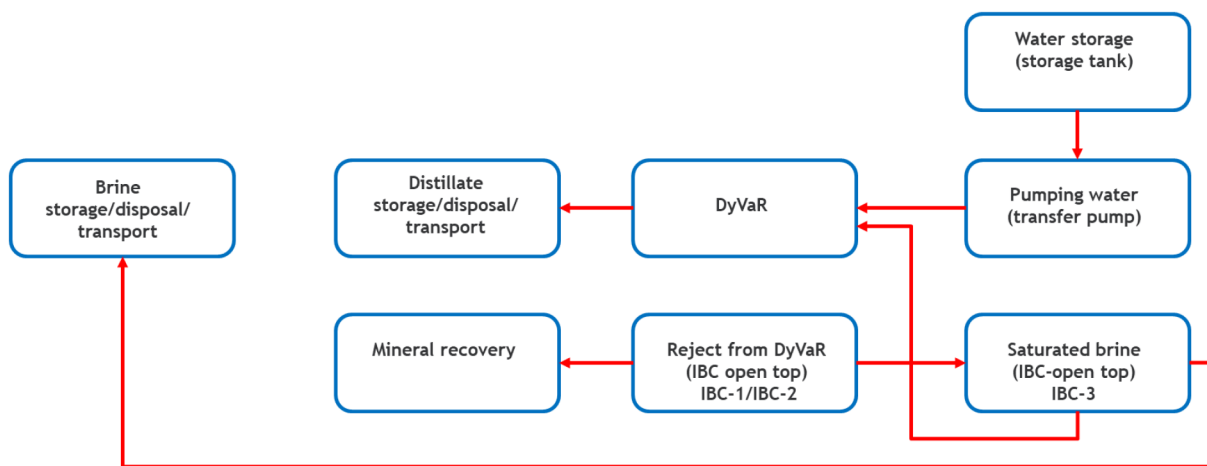


Figure 3—Process flow schematic of DyVaR desalination system

The salt removal process using DyVaR in the ZLD pilot unit consisted of following steps:

1. The clear brine feed (free of H_2S and oil) is transferred from the storage tank using a frequency-controlled transfer pump into the DyVaR unit.
2. The fresh water exits from the DyVaR unit into the distillate storage tank/disposal/transfer.
3. The concentrated salt slurry will be transferred to IBC-1 or IBC-2 which are connected, with one of these IBCs always kept empty.
4. The saturated salt solution from IBC-3 will be pumped back to the DyVaR for processing.
5. IBC-1 or IBC-2 will be removed once filled with salt crystals and flow of salt slurry is directed to the empty IBC-1 or IBC-2.

Results and Discussion

Inlet Feedwater Conditions

Below is a summary of geochemical water analysis results obtained for water samples collected from the WT diverter manifold before sending the water stream into the process tank of the pretreatment unit. The salinities of water samples are significantly high, within range of 247,000 ppm to 257,000 ppm TDS, and they are mainly calcium chloride brines due to the presence of significant amounts of calcium (more than 68,000 ppm) and chloride (about 140,000 ppm) ions. In addition, the feed water samples showed about 1,000 ppm oil in water (OiW) content, more than 200 ppm TSS, and greater than 500 ppm dissolved H_2S .

Table 1—Geochemical water analysis of water from WT diverter manifold

Parameter	Sample #1	Sample #2
TDS (mg/L)	247,000	257,000
TSS (mg/L)	200	220
Chloride/ Cl^- (mg/L)	>140,000	>140,000
Calcium/ Ca^{2+} (mg/L)	>68,000	>70,000

Pretreatment Unit Performance

The pre-treatment module comprising of H_2S scavenging (monoethanolamine/MEA triazine) and N_2 gas floatation was able to successfully reduce the H_2S and OiW to the allowable limit for the DyVaR desalination unit. Figure 4 shows the H_2S content measured in flowback produced water after the pretreatment as well as

the amount of scavenger consumed. A total of ~1,525 bbl of flowback produced water obtained from WT separator was pretreated with an average of 2.9L scavenger consumed per bbl of water processed. All pretreatment runs scavenged H₂S in flowback produced water to less than 1 ppm besides effectively lowering the OiW content to lower than 10 ppm and TSS to below 10 ppm, which confirms the robustness of the pretreatment unit configuration considered for the ZLD pilot.

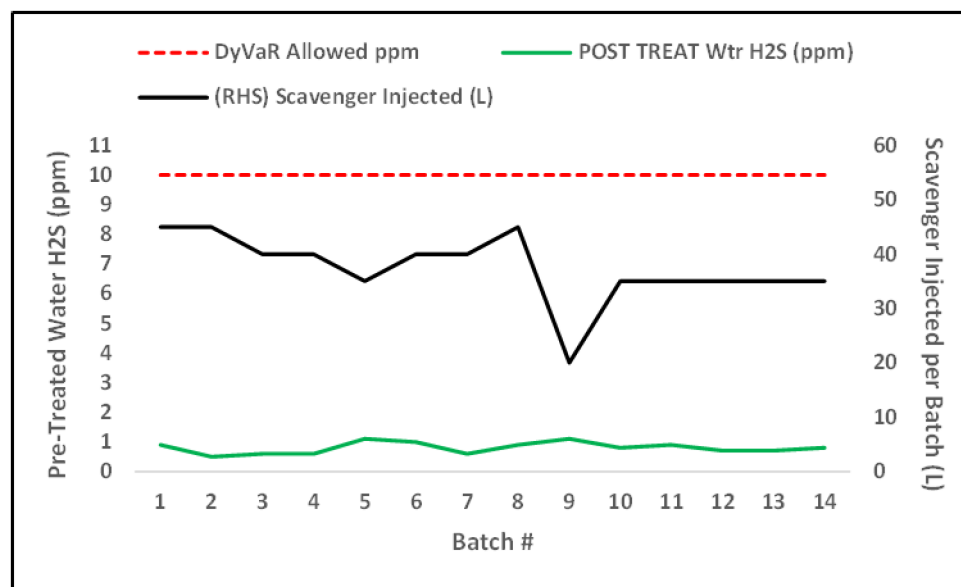


Figure 4—H₂S content in the pretreated water as well as the scavenger consumed

DyVaR Unit Performance

The DyVaR skid was designed to produce a distillate with a target TDS of about 500 ppm and recovery efficiency of 65% to 70%. To monitor the quality of distillate product, regular samples from feed (inlet), distillate, and reject were taken during the trial test period and analyzed for pH, TDS, TSS, conductivity, and turbidity. A range of pretreated flowback produced water compositions and salinities were tested at the DyVaR unit to obtain comprehensive data for optimization of potential upscaling opportunities. The desired range of flowback produced water salinities are obtained by diluting the pretreated water to 20%, 30%, 40%, 50%, 65%, and 80%.

Figure 5 shows the total dissolved solids (TDS) concentration of DyVaR inlet, DyVaR distillate, and DyVaR reject samples for all the batches tested during the pilot operation. The first 14 batches were processed in the DyVaR unit with zero dilution of flowback produced water, 2 batches with 20% dilution (15-16), 4 batches with 30% dilution (17-20), 5 batches with 40% dilution (21-25), 6 batches with 50% dilution (26-31), 1 batch with 65% dilution (32), and the last batch with 80% dilution (33). It can be seen that DyVaR inlet water TDS values match very well with the chosen dilution ratio by resulting in TDS values ranging from 50,000 to 250,000 ppm. The TDS values were effectively lowered from 50,000 ppm to 250,000 ppm in the inlet to less than 500 ppm TDS in the distillate (for all scenarios). Such low TDS obtained for the distillate remains considerably below than the current frac water TDS requirements used. This low TDS performance for the distillate water was consistently achieved during the whole operation duration of the pilot unit. Interestingly salts were effectively concentrated in the DyVaR reject with TDS values reaching as high as 420,000 ppm.

- All DyVaR runs yielded distillate production with TDS less than 500 ppm, with max TDS of 510 ppm and min TDS of 70 ppm

- All DyVaR runs showed that salts were effectively concentrated in the reject with TDS reaching as high as 420,000 ppm.
- While no crystals were found in the reject brine, reject TDS reached above the saturation level.

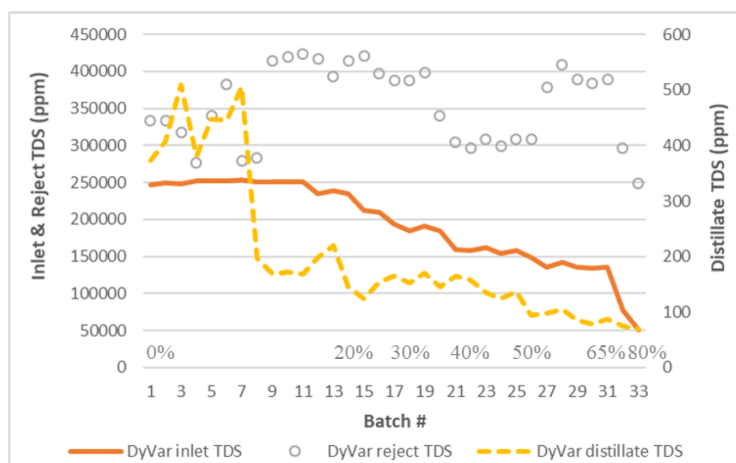


Figure 5—DyVaR Inlet, Distillate & Reject TDS (ppm)

The TSS of distillate water was always found to be less than 5 ppm, turbidity values ranged from 0.4 to 5.6 NTU, and pH was maintained in between 9.1 and 10.5 (Figure 6). Such high pH can be attributed to the presence of H_2S scavenging chemical (monoethanolamine (MEA)-triazine) derivatives in the DyVaR distillate. The excellent quality of distillate water obtained confirms the suitability of DyVaR technology to treat the quite challenging frac flowback produced water streams from unconventional resources with salinities as high as 250,000 ppm TDS. In summary, the consistent performance of the technology to generate low salinity water with a range of flowback produced water salinities (50,000 to 250,000 ppm TDS) indicates that ZLD comprising of a customized pretreatment and DyVaR can potentially become an alternative water resource for frac water applications.

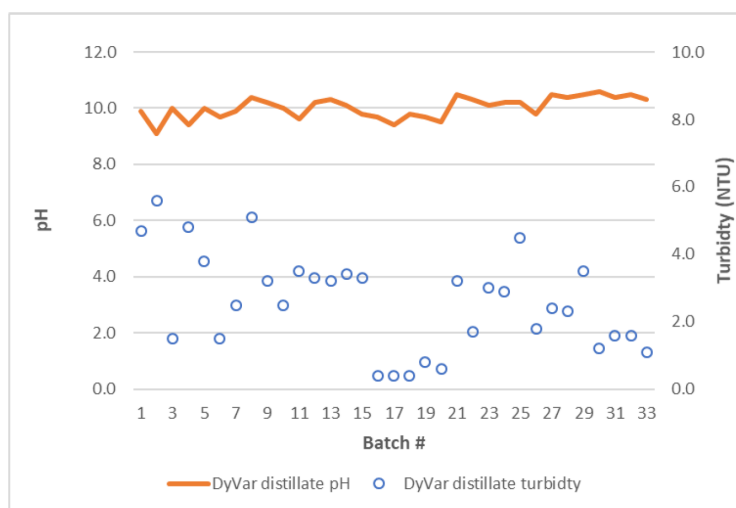


Figure 6—DyVaR distillate pH and turbidity

Prior to the dilution exercise, the DyVaR unit was reaching the internal temperature limit of $\sim 96^\circ\text{C}$, due to the fact that the blower section of the unit is capable of only a maximum 0.5 bara vacuum. Given the supersaturation of the flowback produced water ($\sim 250,000$ ppm), and high concentration of CaCl_2 in the

feed (75%), the system was unable to reach the "sustainable and continuous" operation whereby distillation can occur continuously below the 96°C temperature limit of the unit. Upon 40% dilution, the "sustainable and continuous" condition was reached. As a result, the calculated recovery rate (distillate/feed %) is not representative of the "sustainable and continuous" DyVaR operation for early runs with higher-TDS feed (without dilution). With 40% dilution, the DyVaR unit was run within the "sustainable and continuous" bounds, resulting in representative and consistent 65-70% recovery rates (Figure 7).

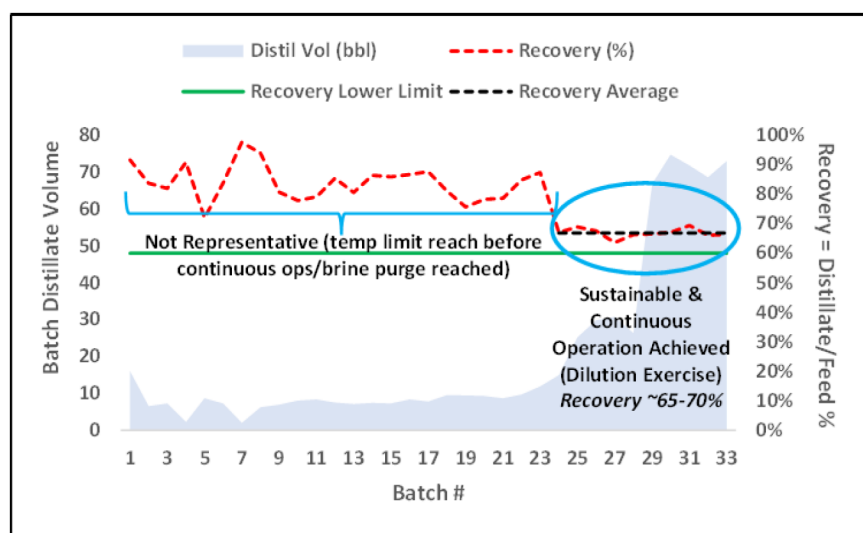


Figure 7—DyVaR distillate volumes and recovery rates

For such high-TDS (and high CaCl₂ concentration) flowback produced water, the inclusion of an additional blower will be able to generate a deeper vacuum and facilitate distillation at a lower mother liquor temperature. At the same time, it is also believed that the target concentration factor (CF) of ~2.5x at ~0.35-0.37 bara vacuum, which results in ~60% water efficiency, could be the optimal set point to minimize energy intensity without driving meaningful excess waste from DyVaR unit.

Generally, an increase in heat exchanger (HTX) pressure drop (DP) serves as evidence of scaling in the DyVaR unit. The pressure drops measured for continuous 6 days of production cycle with 50% diluted flowback produced water are shown in Figure 8. Overall, there was no significant increase in heat exchanger pressure drops during the continuous production cycles. However, the complete analysis identified that a gradual increase in pressure drop was observed with reduced antiscalant dosage. The minimal pressure buildup above 0.60 bar observed across the heat exchanger confirms the antiscalant success criteria. The heat exchanger was also physically opened but no collectable scales were observed (Figure 9). Based on these results, it can be concluded that the small dosages of antiscalant used was able to successfully mitigate scaling issues during the DyVaR operation.

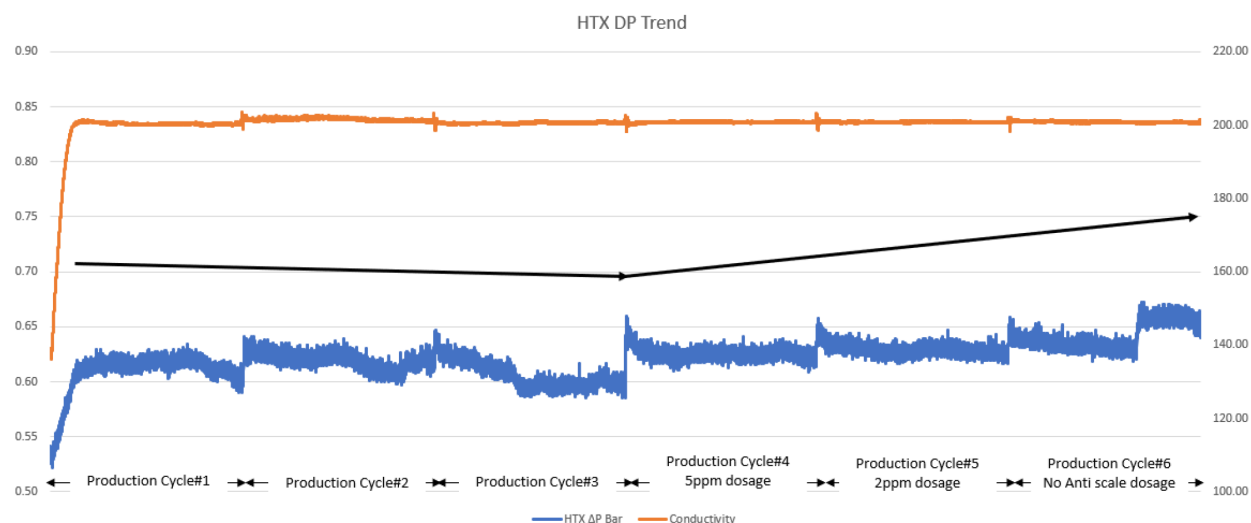


Figure 8—Anti-scalant effect on heat exchanger DP during continuous production cycles



Figure 9—Physical evidence for the absence of scaling inside the heat exchanger

Distillate Water Field Application

Upon the treatment of frac flowback water, a significant volume of distillate treated water was used in unconventional stimulation operations to demonstrate its practicality and operational feasibility. Two wells undergoing hydraulic fracturing operations were selected for a pilot test. The treated water was used in plug-and-perforate (PnP) operations, used to lower perforating guns inside the well, carried by pumped fluids, typically freshwater. Out of the two wells, pump down operations in 12 different stages were completed successfully. Pumping rates were maintained at the desired values, with no complications on pumping pressure. The repeatability of these findings across all 12 stages demonstrated the robustness of distillate water in hydraulic fracturing operations, proving it as a suitable alternative to freshwater. Moreover, lab testing was carried out to quantify the differences in viscosity performance with the addition of friction reducers (FR). The distillate water sample exhibited higher viscosities at the same FR concentration compared to conventional water samples. This highlights that distillate water had more friction reduction capabilities at the same concentration, which infers potential to lower FR consumption in unconventional stimulation operations, further underscoring its contribution to sustainability.

Summary and Conclusions

In this paper, we presented the data collected from a successful ZLD pilot execution to treat the quite challenging extreme salinity unconventional frac flowback produced water. The pilot performance was

evaluated in terms achieving the success key performance indicators (KPIs) to provide useful guidance for the full-scale optimization. Overall, the pilot showed good results for 1) pre-treatment effectiveness, 2) distillate quality, 3) continuous operations, and 4) antiscalant performance, over a range of feed water concentrations (dilution up to 80%), which provides a comprehensive assessment to optimize ZLD design based on location-specific water qualities for different unconventional fields. The major findings are summarized below in the following.

- *Pretreatment effectiveness*: The pretreatment unit composed of H₂S scavenging and N₂ gas flotation was able to lower the H₂S content of the flowback produced water from an initial >500 ppm to less than 1 ppm, much lower than a designed target value of below 10 ppm. Overall, about 2.93 liters of scavenger were consumed per bbl of processed water during the pretreatment. In addition, TSS was reduced to below 10 ppm and OiW content values from 1,000 ppm to less than 10 ppm.
- *Distillate quality (TDS)*: The distillate product water obtained from pilot unit is a clear water with less than 500 ppm TDS and without any appearance of suspended solids. This excellent quality characteristic of the distillate product remained consistent throughout the pilot period. The performance of the technology far exceeds the desired frac water qualities in terms of the TDS content.
- *Recovery rate (waste minimization)*: The pilot system was able to run within the "sustainable and continuous" bounds of the chosen DyVaR unit using the dilution exercise. A representative 65-70% recovery rate was observed at dilution rates above 40%, which meets the targeted KPI range. The testing with an advanced DyVaR pilot system capable of a deeper 0.3bara vacuum was recommended to evaluate whether the "sustainable and continuous" operation could be reached on the raw (non-diluted) extreme flowback produced water of 250,000ppm.
- *Continuous operations*: The continuous operation feature of ZLD pilot was demonstrated at 40%, 50%, 60%, and 80% dilution ratios, which supports the continuous operation capability of the DyVaR ZLD unit. Such successful testing in continuous operation mode meets one set KPI of the pilot by gathering all the data necessary to demonstrate the continuous operability at full scale.
- *Anti-scaling performance*: Small dosages of anti-scaling chemical used was able to successfully mitigate scaling issues across the heat exchanger of DyVaR unit to minimize OPEX and optimize recovery. This can be considered as a major success KPI considering the high concentration of scaling agents in the flowback produced water.
- *Treated water field application*: Treated distilled water from the DyVar unit was successfully utilized in plug-and-perforate pump down operations across 12 hydraulic fracturing stages in two separate wells. The deployment yielded comparable pumping rates and pressures to those achieved with freshwater, with no operational challenges encountered. Viscosity performance lab testing indicated that distilled water exhibited higher viscosity at the same FR concentration compared to typical water samples. This infers that utilizing distilled water in hydraulic fracturing operations could potentially reduce the amount of FR required, leading to more efficient and sustainable operations.

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